

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T32 · Linear Graphs $y = mx + c$

With Answers

Questions with full worked solutions

■ 41 pages

FAMILY MAP

Linear Graphs $y = mx + c$

F01	Construct a straight-line equation	8 Q 30 marks
F02	Read gradient and intercept structure	8 Q 12 marks
F03	Generate and plot a linear graph	2 Q 10 marks

Working Unit 15

4MA1-1H Q25

ORIGINAL QUESTION

5 marks

25 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates $(4, 7)$

R is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points Q and S

Give your answer in the form $ay = bx + c$ where a , b and c are integers.

(Total for Question 25 is 5 marks)

Worked Solution 15

Equation of the other diagonal of a square

UNIT 1
5 marks**Midpoint**

$$M_{PR} = \left(\frac{4+8}{2}, \frac{7-5}{2} \right) = (6, 1).$$

Perpendicular gradient

$$m_{PR} = \frac{-5-7}{8-4} = -3 \Rightarrow m_{QS} = \frac{1}{3}.$$

Equation through the midpoint

$$y - 1 = \frac{1}{3}(x - 6) \Rightarrow y = \frac{1}{3}x - 1 \Rightarrow 3y = x - 3.$$

Final answer

$$3y = x - 3$$

Dependency check Square diagonals bisect each other at right angles.

Question 25

4MA1-2H Q18

ORIGINAL QUESTION

2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

Worked Solution 25

Perpendicular gradient

MODULAR UNIT 1

2 marks

Gradient

$$5y + 2x = 7 \Rightarrow m = -2/5; \quad m_{\perp} = 5/2.$$

Final answer

$$\frac{5}{2}$$

Question 21

4MA1-2H Q9 whole

ORIGINAL QUESTION UNIT 1

2 marks

- 9 A straight line passes through the points with coordinates $(0, -3)$ and $(2, 0)$

Find an equation of the line.

.....

Worked solution 21

Equation of a straight line

MODULAR UNIT 1

2 marks

whole – Equation of a straight line

Find the gradient

$$m = \frac{0 - (-3)}{2 - 0} = \frac{3}{2} = 1.5.$$

Use the y-intercept

The point $(0, -3)$ gives $c = -3$, so $y = 1.5x - 3$.**Final answer**

$$y = 1.5x - 3$$

Question 06

4MA1-1H Q13

ORIGINAL QUESTION

3 marks

13 A is the point with coordinates $(-5, 12)$

B is the point with coordinates $(19, -48)$

Find an equation of the straight line that passes through the points A and B

Worked Solution 06

Equation of a line through two points

MODULAR UNIT 1

3 marks

Find the gradient

$$m = \frac{-48 - 12}{19 - (-5)} = \frac{-60}{24} = -2.5.$$

Find the intercept

$$12 = -2.5(-5) + c \Rightarrow c = -0.5.$$

Final answer

$$y = -2.5x - 0.5$$

Original question 25

4MA1-2H Q18 M01

MODULAR UNIT 1

4 marks

- 18 The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates (8, 11)

Find an equation for **L**

Give your answer in the form $y = mx + c$

.....

Worked solution 25

Find a perpendicular line

MODULAR UNIT 1

4 marks

M01 – Find a perpendicular line**Read the original gradient** $2x + y = 9 \Rightarrow y = -2x + 9$, so $m = -2$.**Use the negative reciprocal**

$$m_{\perp} = 1/2.$$

Use (8,11)

$$11 = \frac{1}{2}(8) + c \Rightarrow c = 7.$$

State the line

$$y = \frac{1}{2}x + 7.$$

Final answer

$$y = \frac{1}{2}x + 7$$

Original question 08

4MA1-1H Q10 Q10(a)

ORIGINAL QUESTION

2 marks

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

(2)

Worked solution 08Chapter: Linear Graphs ($y = mx + c$)

MODULAR UNIT 1

2 marks

Q10(a) – Chapter: Linear Graphs ($y = mx + c$)Family: Read gradient and intercept structure • Variant: Gradient and intercept from $y=mx+c$ **Method** $2y+7x=10$ gives $2y=10-7x$.*Why: Rearrange into $y=mx+c$ form.***Method** $y=5-3.5x$, so the gradient is -3.5 .*Why: The coefficient of x is the gradient.***Final answer**gradient: -3.5

Original question 09

4MA1-1H Q10 Q10(b)

ORIGINAL QUESTION**1 marks**

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

(b) Find the coordinates of the point where **L** crosses the y -axis.

(.....,)
(1)

Worked solution 09Chapter: Linear Graphs ($y = mx + c$)

MODULAR UNIT 1

1 marks

Q10(b) – Chapter: Linear Graphs ($y = mx + c$)Family: Read gradient and intercept structure • Variant: Gradient and intercept from $y=mx+c$ **Method**At the y-axis $x=0$, so $2y=10$ and $y=5$; the intercept is $(0,5)$.*Why: Set x equal to zero to find the y-intercept.***Final answer****y-intercept:** $(0,5)$

Line **L** has equation $y = 2 - 3x$

(b) Write down the gradient of line **L**.

(1)

Worked solution

January 2020 | Question 1 | 1 marks

Family: Read gradient and intercept structure • Variant: Gradient and intercept from $y=mx+c$

Method / working

1. In $y=2-3x$, the coefficient of x is the gradient.

Final answer: -3

Line **L** has equation $y = 2 - 3x$

- (c) Does the point with coordinates $(100, -302)$ lie on line **L**?
You must give a reason for your answer.

(1)

Worked solution

January 2020 | Question 1 | 1 marks

Family: Read gradient and intercept structure • Variant: Gradient and intercept from $y=mx+c$

Method / working

1. Substitution gives -298, so (100,-302) is not on the line.

Final answer: No; $2-3(100)=-298 \neq -302$

19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

(1)

Worked solution

January 2020 | Question 19 | 1 marks

Family: Construct a straight-line equation • Variant: Parallel line through a point

Method / working

1. A parallel line has the same gradient as $y=7-4x$; using the official diagram convention the required parallel gradient is $+4$, so $y=4x$ is valid.

Final answer: $y=4x$

The line **L** passes through the points with coordinates $(-3, 1)$ and $(2, -2)$

- (b) Find an equation of the line that is perpendicular to **L** and passes through the point with coordinates $(-6, 4)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

.....
(4)

Worked solution

January 2020 | Question 19 | 4 marks

Family: Construct a straight-line equation • Variant: Perpendicular line through a point or midpoint

Method / working

1. Gradient of L through $(-3,1), (2,-2)$ is $-3/5$.
2. A perpendicular gradient is $5/3$.
3. Through $(-6,4)$: $y-4=(5/3)(x+6)$.
4. Multiply by 3 and rearrange: $5x-3y+42=0$.

Final answer: $5x-3y+42=0$

- 13** Point A has coordinates $(5, 8)$
Point B has coordinates $(9, -4)$
- (a) Work out the gradient of AB .

(2)

Worked solution

January 2021 | Question 13 | 2 marks

Family: Read gradient and intercept structure • Variant: Gradient from two points or a graph

Method / working

1. Gradient= $(-4-8)/(9-5)=-12/4$.

2. Therefore the gradient is -3.

Final answer: -3

The straight line **L** has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to **L**.

(1)

Worked solution

January 2021 | Question 13 | 1 marks

Family: Read gradient and intercept structure • Variant: Parallel and perpendicular gradient relation

Method / working

1. A perpendicular gradient is the negative reciprocal of -4 , namely $\frac{1}{4}$.

Final answer: $\frac{1}{4}$

17 The straight line **L** passes through the points $(4, -1)$ and $(6, 4)$

The straight line **M** is perpendicular to **L** and intersects the y -axis at the point $(0, 8)$

Find the coordinates of the point where **M** intersects the x -axis.

(.....,)

Worked solution

June 2021 | Question 17 | 4 marks

Family: Construct a straight-line equation • Variant: Perpendicular line through a point or midpoint

Method / working

1. Gradient of $L = (4 - (-1)) / (6 - 4) = 5/2$.
2. Perpendicular gradient of $M = -2/5$.
3. Use $y = -(2/5)x + 8$ and set $y = 0$.
4. Solve $x = 20$, so the x-intercept is $(20, 0)$.

Final answer: $(20, 0)$

- 20** The straight line **L** passes through point $A(-6, 2)$ and point $B(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B .
The line **M** intersects the line $x = -1$ at point C .

Calculate the area of triangle ABC .

Worked solution

November 2021 | Question 20 | 7 marks

Family: Construct a straight-line equation | Variant: Perpendicular line through a point or midpoint

METHOD / WORKING

1. $A=(-6,2), B=(5,3)$

2. The midpoint of AB is

3. $\left(\frac{-6+5}{2}, \frac{2+3}{2}\right)$

4. $=$

5. $\left(-\frac{1}{2}, \frac{5}{2}\right)$

6. The gradient of AB is

7. $\frac{3-2}{5-(-6)} = \frac{1}{11}$

8. So the perpendicular gradient is

9. -11

10. Line M passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

11. $y - \frac{5}{2} = -11\left(x + \frac{1}{2}\right)$

12. At point C , $x = -1$:

13. $y - \frac{5}{2} = -11\left(-\frac{1}{2}\right)$

14. $y - \frac{5}{2} = \frac{11}{2}$

15. $y=8$

16. So

17. $C=(-1,8)$

18. Now use vectors from A :

19. $\vec{AB}=(11,1), \vec{AC}=(5,6)$

20. Area of triangle ABC is

21. $\frac{1}{2} |11(6)-1(5)|$

22. $=\frac{1}{2}(61)$

23. $=30.5$

FINAL ANSWER

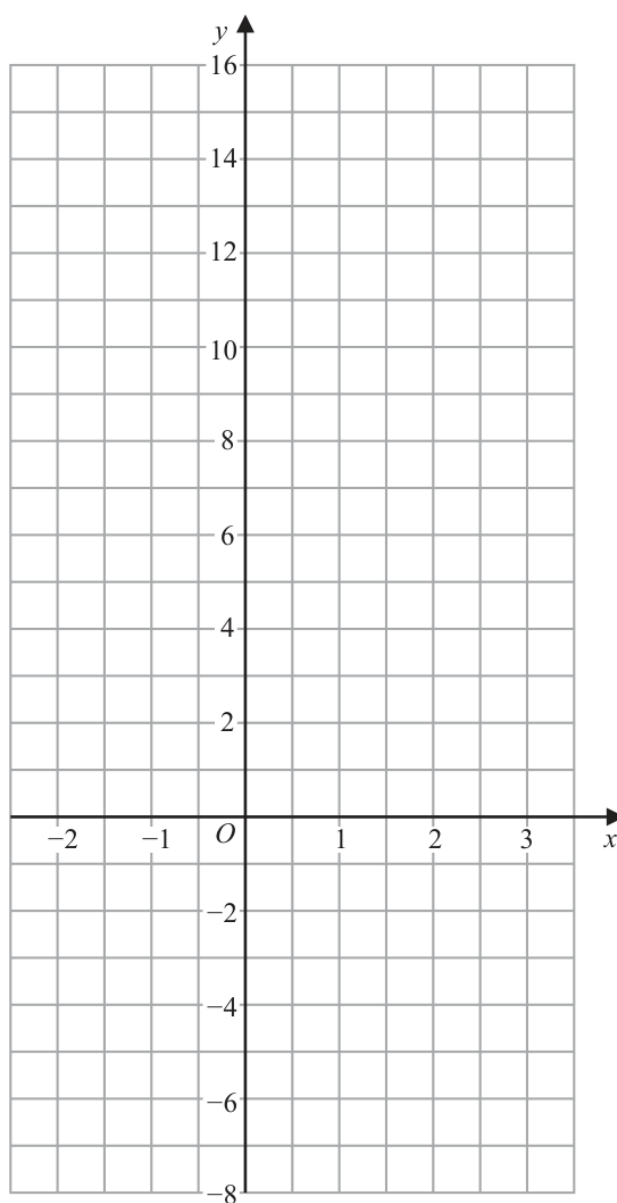
Worked solution

METHOD / WORKING

Worked solution (continued)
24. Answer: 30.5

FINAL ANSWER	30.5
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- 3 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3



(Total for Question 3 is 3 marks)

Answer reference | May-Nov 2020 | Question 3

Family: Generate and plot a linear graph • Variant: Plot a line from generated values

3	<table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>15</td><td>11</td><td>7</td><td>3</td><td>-1</td><td>-5</td></tr></table>	x	-2	-1	0	1	2	3	y	15	11	7	3	-1	-5	Correct line between $x = -2$ and $x = 3$	3	B3 for a correct line between $x = -2$ and $x = 3$ (B2 for a correct straight line segment through at least 3 of $(-2, 15)$ $(-1, 11)$ $(0, 7)$ $(1, 3)$ $(2, -1)$ $(3, -5)$ or for all of $(-2, 15)$ $(-1, 11)$ $(0, 7)$ $(1, 3)$ $(2, -1)$ $(3, -5)$ plotted but not joined) (B1 for at least 2 correct points stated (may be in a table) or plotted or for a line drawn with a negative gradient through $(0, 7)$ or for a line with a gradient of -4)
	x	-2	-1	0	1	2	3											
y	15	11	7	3	-1	-5												
$(-2, 15)$ $(-1, 11)$ $(0, 7)$ $(1, 3)$ $(2, -1)$ $(3, -5)$																		
				Total 3 marks														

Worked solution

May-Nov 2020 | Question 3 | 3 marks

Family: Generate and plot a linear graph • Variant: Plot a line from generated values

Solution

$$y=7-4x$$

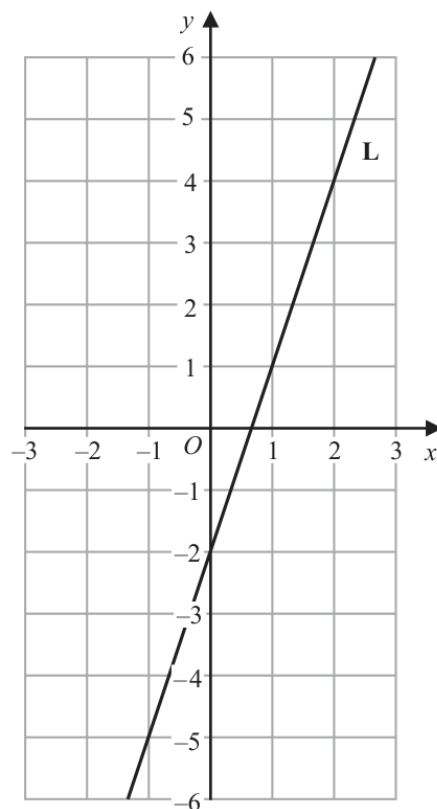
Useful points are:

$(-2,15)$, quad $(0,7)$, quad $(1,3)$, quad $(2,-1)$, quad $(3,-5)$

Plot these points and join them with a straight line.

Answer: The graph is the straight line $y=7-4x$.

- 6 The line **L** is shown on the grid.



Find an equation for **L**.

(Total for Question 6 is 2 marks)

Answer reference | May-Nov 2020 | Question 6

Family: Read gradient and intercept structure • Variant: Parallel and perpendicular gradient relation

6				M1 for $y = 3x + c$ oe or $y = mx - 2$ oe or $3x - 2$ or eg $L = 3x - 2$ or $y = 3(x \pm a)$
		$y = 3x - 2$	2	A1 oe eg $y - 4 = 3(x - 2)$ $y - 1 = 3(x - 1)$ $y - a = 3(x - b)$ where (a, b) is any coordinate on the line
				Total 2 marks

Worked solution

May-Nov 2020 | Question 6 | 2 marks

Family: Read gradient and intercept structure • Variant: Parallel and perpendicular gradient relation

Official final mark-scheme pages are embedded immediately after this question.

- 9 (a) Write down the value of $(m + 2)^0$ where m is a positive integer.

.....
(1)

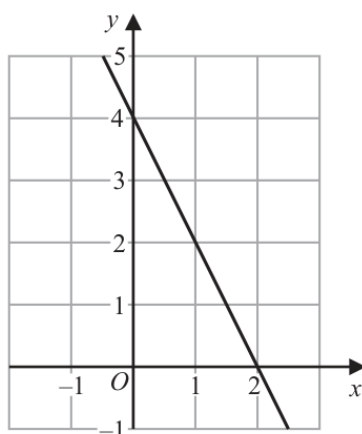
- (b) Simplify $(3a^2b^4)^3$

.....
(2)

- (c) Factorise fully $14x^2y^4 + 21x^3y^2$

.....
(2)

The diagram shows a straight line drawn on a grid.



- (d) Write down an equation of the line.

.....
(2)

(Total for Question 9 is 7 marks)

Answer reference | January 2023 | Question 9

Family: Generate and plot a linear graph • Variant: Plot a line using axis intercepts

Question	working	Answer	Mark	Notes
9 (a)		1	1	B1
(b)		$27a^6b^{12}$	2	B2 (B1 for 2 of 3 parts in a product)
(c)		$7x^2y^2(2y^2+3x)$	2	B2 B1 for a correct factorisation with at least 2 factors outside (eg $7x, x^2, xy$, etc) eg $7x(2xy^4+3x^2y^2)$ eg $x^2y^2(14y^2+21x)$ or for the correct common factor with just one mistake inside the bracket eg $7x^2y^2(2y+3x)$ which is missing the squared on the y term
(d)	$y = mx + 4$ where $m \neq 0$ oe (eg $y = 2x + 4$) or $y = -2x + c$ or $y + 2x = c$ oe or $-2x + 4$ or $f(x) = -2x + 4$ oe		2	M1
	Correct answer scores full marks (unless from obvious incorrect working)	$y = -2x + 4$		A1 oe eg $y + 2x = 4$
				Total 7 marks

Worked solution

January 2023 | Question 9 | 7 marks

Family: Generate and plot a linear graph • Variant: Plot a line using axis intercepts

Official final mark-scheme pages are embedded immediately after this question.